



12582 - Building the Legacy of Supernova 2023ixf: How Does Molecule Formation Lead to Dust?

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

Folder	Observation	Label	Observing Template	Science Target
1500d				
	7	NIRSpec	NIRSpec Fixed Slit Spectroscopy	(1) SN2023ixf
	8	MIRI LRS	MIRI Low Resolution Spectroscopy	(1) SN2023ixf
	9	MIRI Imaging	MIRI Imaging	(1) SN2023ixf

ABSTRACT

Core collapse supernovae (SNe) with hydrogen-rich envelopes, aka Type II SNe (SNe II), are the most common stellar explosions in the Universe. They are the main producers of heavy elements, and produce significant amounts of dust. Thus, the study of these cosmic explosions probes the chemical evolution of the universe, sheds light on the composition of dust in our solar system, and ultimately the genesis of life. Despite the prevalence of SNe II, their exact role as dust producers is poorly understood. Molecules (e.g. CO and SiO) have been observed in the in the ejecta that form dust. These molecules determine which elements are present in interstellar gas and which are tied up in cosmic dust. To make progress in understanding the role of SNe II in dust formation, we must observe nearby SN from days to years after explosion. Here, we request 20.0 hours of time to observe the closest SN II in JWSTs lifetime, SN 2023ixf. Data will be obtained at four epochs from 500-1500 d past explosion. Allowing us to determine, i) How molecule formation leads to dust formation? and ii) When, what kind, and how much dust is produced in the ejecta? With a

spectroscopic time series of data, for the first time, we can use molecules to trace the physical conditions throughout the epochs of dust formation, along with the composition of the dust. JWST spectra will reveal the pathway from molecules to dust in a way which was not previously possible. Finally, the data presented here will contribute an important part of the SN 2023ixf legacy dataset which will be used for decades to come.

OBSERVING DESCRIPTION

We will obtain four epochs of data of the nearby hydrogen rich type II supernova 2023ixf, ranging from ~2-25 micron. These data will be observed over the next three cycles at the following dates and SN phases from explosion:

- 1) 6th December 2024 - 8th January 2025; +567-600 d,
- 2) 18th April 2025 - 7th June 2025; +700-750 d,
- 3) 18th April 2026 - 7th June 2026; +1065-1165 d,
- 4) 18th April 2027 - 7th June 2027; +1430-1530 d.

Our science case requires spectra, hence we will use NIRSpec (395M/F290LP+G235M/F170LP) and MIRI/LRS. To complete the SED at longer wavelengths we obtain F1500W, F1800W, F2100W and F2550W imaging. In total this requires 20.0hr of charged time

Proposal 12582 - Targets - Building the Legacy of Supernova 2023ixf: How Does Molecule Formation Lead to Dust?

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	SN2023ixf	RA: 14 03 38.5640 (210.9106833d) Dec: +54 18 42.02 (54.31167d) Equinox: J2000		
Comments: Category=Star Description=[Type II supernovae] Extended=NO					

Proposal 12582 - Observation 7 - Building the Legacy of Supernova 2023ixf: How Does Molecule Formation Lead to Dust?

Observation	Proposal 12582, Observation 7: NIRSpec										Wed Apr 15 21:01:34 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	SN2023ixf	RA: 14 03 38.5640 (210.9106833d) Dec: +54 18 42.02 (54.31167d) Equinox: J2000									
	Comments: Category=Star Description=[Type II supernovae] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	1 SN2023ixf	WATA	SUB2048	CLEAR	NRSRAPID	3	1	1	3.628	173508	
Template	HFF Readout Mode				Slit			Subarray				
	false				S400A1			SUBS400A1				
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395M/F290LP	S400A1	NRS	20	3	1	NONE	3	9	1135.966	
	2	G235M/F170LP	S400A1	NRS	26	3	2	NONE	3	9	1472.494	

Special Requirements	Between Dates 18-APR-2027:00:00:00 and 01-JUN-2027:00:00:00
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Proposal 12582 - Observation 8 - Building the Legacy of Supernova 2023ixf: How Does Molecule Formation Lead to Dust?

Observation	Proposal 12582, Observation 8: MIRI LRS									Wed Apr 15 21:01:34 GMT 2026
	Diagnostic Status: Warning									
Diagnostics	Observing Template: MIRI Low Resolution Spectroscopy									
	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(1)	SN2023ixf	RA: 14 03 38.5640 (210.9106833d) Dec: +54 18 42.02 (54.31167d) Equinox: J2000							
Acquisition	Comments: Category=Star Description=[Type II supernovae] Extended=NO									
	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
Template	1	1 SN2023ixf	F1000W	FASTGRPAVG	4	1	1	44.401	173508	
	Subarray				Obtain Verification Image?					
Dithers	FULL				true					
	#	Dither Type	No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset	
Pointing Verification	1	ALONG SLIT NOD								
	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	10	4	4	1	1	119.327		F770W

Proposal 12582 - Observation 8 - Building the Legacy of Supernova 2023ixf: How Does Molecule Formation Lead to Dust?

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	100	3	6	1	2	1676.124	
Special Requirements	Between Dates 18-APR-2027:00:00:00 and 01-JUN-2027:00:00:00								

Proposal 12582 - Observation 9 - Building the Legacy of Supernova 2023ixf: How Does Molecule Formation Lead to Dust?

Observation	Proposal 12582, Observation 9: MIRI Imaging										Wed Apr 15 21:01:34 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	SN2023ixf	RA: 14 03 38.5640 (210.9106833d)								
			Dec: +54 18 42.02 (54.31167d)								
			Equinox: J2000								
Template	Comments: Category=Star Description=[Type II supernovae] Extended=NO										
	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets	1	10		1	1	POINT SOURCE	POSITIVE	DEFAULT	
	2	CYCLING	1	10						DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F1500W	FASTR1	10	2	1	Dither 1	4	8	233.103	173508
	2	F1800W	FASTR1	20	2	1	Dither 1	4	8	455.107	173508
	3	F2100W	FASTR1	30	2	1	Dither 1	4	8	677.11	173508
	4	F2550W	FASTR1	22	10	1	Dither 2	10	100	6354.842	173508
Special Requirements	Between Dates 18-APR-2027:00:00:00 and 01-JUN-2027:00:00:00										